

Case Study of a Model that evaluates the Learner Experience with DICTs

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ABSTRACT

Learner eXperience (LX) can be defined as the perceptions, answers, and performances of learners interacting with learning environments, educational products, and resources. Evaluating the LX to obtain experiences that support and facilitate learning and knowledge mastery is important. Thus, we developed the Learner Experience Evaluation Model (LEEM) to assess and improve the learner's experience using Digital Information and Communication Technologies (DICTs) during learning. Therefore, this paper presents a case study to evaluate the applicability of the LEEM's checklists in a real learning context. 23 learners and 1 teacher of the Software Engineering discipline participated in this study. The results of the study were analyzed quantitatively and qualitatively. Through the LEEM application, the teacher noticed that the learners felt comfortable using the DICTs and that they were easy to use. This study contributes to evaluating the LX at different moments of educational activity in a real context using LEEM.

CCS CONCEPTS

• **Applied computing** → *Education*; • **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Learner eXperience, Evaluation, Model, LX, Qualitative Analysis, Quantitative Analysis, Student eXperience

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1 INTRODUCTION

Digital Information and Communication and Technologies (DICTs) is gradually being inserted into human life to facilitate and assist in their daily activities, such as Internet of Things (IoT), Cloud computing, among others. When applied to the school environment, DICTs can provide interactive and collaborative experiences [1, 6, 14]. Therefore, it is important to evaluate these educational experiences that use DICTs, as a way of verifying their contributions and impacts on learning [5].

In this sense, the term Learner eXperience (LX) arises, which is a concept derived from User eXperience (UX) [8]. The definition of UX according to ISO 9241-210 [7] is user perceptions and responses that result from the use or anticipated use of a system, product, or service. LX, according to Huang et al. [5], is defined as learners' perceptions, responses, and performances through interaction with educational technologies. In addition, Shi [22] conceptualizes LX as a learner's perceptions and responses, which involves learner behavior, attitudes, beliefs, sensations, and emotional responses obtained, among others. Therefore, there is a need to create experiences that facilitate learning and knowledge mastery [4] and support in the learning experiences in activities that are out of their reach, seeking protagonism in the learning process [23]. In this sense, Soloway et al. [23] introduces the term Learner, which is used to refer to students who are constantly learning or to professionals who put themselves in a position to learn, subjecting themselves to all the challenges faced when using DICTs.

Huang et al. [5] believe that LX with DICTs can be evaluated and improved through different elements, such as Value, Usability, Adaptability, Comfortability, and Desirability. The elements of LX refer to the components that guide the process of evaluating LX, making it possible to verify various characteristics of the experience, including feelings and emotions in learning. In this way, the elements of LX can vary depending on the evaluation goal, the type of artifact chosen, and the learning theory adopted. Huang et al. [5] recommend that LX be evaluated comprehensively, taking into account the diversity of learners and their learning preferences, so that all aspects of the experiences are evaluated.

Based on this need, the Learner Experience Evaluation Model (LEEM) was proposed, which evaluates LX with the use of DICTs. It has three evaluation stages: a pre-evaluation, an during-evaluation,

and a post-evaluation of LX. The LEEM consists of three checklists and a set of questions. The LEEM enables continuous monitoring and recording of LX progress. At each stage it is possible to use different ways of assessing LX, thus generating the possibility of assessing various elements of LX. Therefore, a case study was carried out to evaluate LX in a real context using LEEM. The research question of this study is: “Does LEEM support the evaluation and improvement of LX during a real educational activity using DICTs?”. 23 learners and one lecturer from the Software Engineering (SE) course in Computer Science and Biomedical Informatics (CS & BI) at the Federal University of Paraná (UFPR) in Brazil participated in this study. The participants’ answers obtained through the LEEM checklists and set of open questions were analyzed quantitatively and qualitatively. The results showed that for some learners, the educational activity generated some challenges, such as managing a project and greater dedication to finding materials. For others, the activity left them in their comfort zone, that is, it didn’t generate any new challenges.

In general, the study presented in this paper contributes to the area of Human-Computer Interaction (HCI), as it evaluates LX using DICTs through LEEM. Understanding LX during the learning process is fundamental, as it provides teachers with insights into the importance of reviewing and, if necessary, adapting the DICTs used, according to the individual needs of the learners [13]. This is also important to avoid situations where learners do not feel comfortable with DICTs, promoting a more effective interaction and a more positive LX [5]. In addition, LEEM can help teachers rethink their teaching and learning strategies when they notice that learners have reported difficulties with the resources adopted. Finally, teachers can also observe whether learners remain motivated in the educational activity and what could be improved.

2 RELATED WORKS

In Magyar and Haley [12], the process of balancing LX and UX took place when designing and developing a peer feedback web application for Massive Open Online Courses (MOOCs), called Gallery Tool. The tool is learner-centered, so it provides a space for sharing work and access to peer feedback. A case study was carried out, in which learners tested the tool in three MOOCs. In addition, they participated in an interview that investigated their motivation for using the tool, what they thought about the peer feedback components and the impact of their experience. The Value and Usability elements were evaluated through questions, such as “What impact it may have had on their MOOC experience?”. Overall, the Gallery Tool had a neutral impact on the learners’ experience. When asked how the tool affected their experience of the course, learners said they had no interest in using it. Others said the tool had a positive effect on them because it motivated, challenged, or inspired them.

Ruiz and Snoeck [19] investigated learners’ experiences when creating a user interface model by adapting Kirkpatrick’s model using the Feedback-Enabled Interface Simulation (FENiKS) tool. The LX elements are evaluated using the metrics of the Kirkpatrick model, namely: Reaction, Learning, Behavior, and Results. Through the adaptation of Kirkpatrick’s model using FENiKS, important insights were gained, on how to what extent the FENiKS tool helped

the learners in learning design. When asked if they had a positive learning experience, the majority of learners agreed with the sentence. Overall, in this case study, it was clear that FENiKS was useful for the learners’ learning.

Martinelli and Zaina [13] investigated the experiences of learners in the context of a Flipped Classroom in online teaching during the COVID-19 pandemic. The authors proposed evaluating the LX both throughout the learning process and at the end of educational activities. In this work, a questionnaire and interview were used. LX was assessed based on the elements of Satisfaction, Motivation, and Feeling towards learning. Through this work, it was possible to understand the human factors that influence the experience of learners, especially in remote classes [13]. Overall, the findings showed that learners felt satisfied and motivated with the materials and gave more positive than negative feedback.

The works cited have limitations. For example, in the Magyar and Haley [12] work, learners didn’t use the tool as expected. Instead of receiving feedback from peers, learners used the Gallery Tool to finish the course or as a source of inspiration for their task. In Ruiz and Snoeck [19]’s work, evaluation results are limited to data collected only during training. The ideal scenario would be that after training, learners continued using the tool for a longer period and provided feedback. In Martinelli and Zaina [13]’s work, as the questionnaire data was anonymous, it was not possible to identify whether the learner’s profile influenced the learner’s experiences. In LX evaluation, the ideal is for the evaluation to occur longitudinally, that is, before, during, and after an educational activity [15]. A longitudinal evaluation allows for continuous monitoring, recording the progress of the learner’s experience and analyzing the relationship between the learner’s experience and their profile. Additionally, it is important to evaluate LX in different ways and consider different elements, as this allows you to evaluate LX more effectively [5]. A model that met all the characteristics mentioned above was not identified in the literature. To cover this gap, the LEEM was proposed.

3 LEARNER EXPERIENCE EVALUATION MODEL (LEEM)

LEEM was organized into three stages of evaluation to allow continuous monitoring and recording of LX’s progress during an educational activity, as suggested by Nygren et al. [15]. The first stage, called pre-evaluation, aims to identify the learners’ profile, considering their difficulties and needs regarding their learning experience, as suggested by Lykke et al. [11]. Pre-evaluation can help form groups for collaborative activities, whether at school or university. It is suggested that the teacher apply group formation techniques according to their familiarity. Twelve pairs of words were created for the LEEM model’s pre-evaluation checklist. For example (Figure 1), in the item “I prefer to work in a way”, the learner can select the option closest to the word they want, “practical” or “theoretical”.

In the 2nd stage, called during-evaluation, LX evaluation takes place during the educational activity. This evaluation seeks to record the feelings and information of the learning experiences from both the learner’s and the teacher’s point of view, as suggested by Nygren et al. [15]. For this reason, the checklists that the learner and teacher will answer have similar sentences, to assess both their perspectives

I prefer to work like this:

PRACTICE  THEORETICAL

When I am take responsibility, I feel:

OPTIMIST  PESSIMIST

Figure 1: Part of the LEEM pre-evaluation checklist

on the learning process. The learner will report their feelings using the Self-Assessment Manikin (SAM) scale [10], such as about finding support material (Figure 2). The learner will also report their record of the learning experience using a 5-point Likert scale about their preference for using DICTs for learning. The teacher, meanwhile, will assess on a 5-point Likert scale whether they agree that the learner enjoyed accessing the support material (Figure 3). It is recommended to apply these checklists after 50% of the educational activity because both the learner and the teacher will already have had some experience to give their feedback. For example, if the educational activity lasts two weeks, then these checklists should be applied at the end of the first week.

How did you feel when looking for support material?

Totally Sad

Totally Happy

NOT APPLICABLE

How did you feel working collectively?

Totally Sad

Totally Happy

NOT APPLICABLE

Figure 2: Part of the LEEM during-evaluation checklist (learner)

I noticed that the student(s) enjoyed accessing support material.

1 2 3 4 5

TOTALLY DISAGREE TOTALLY AGREE

I noticed that the student(s) enjoyed working collectively.

1 2 3 4 5

TOTALLY DISAGREE TOTALLY AGREE

Figure 3: Part of the LEEM during-evaluation checklist (teacher)

The third and final stage, called post-evaluation, aims to evaluate the learner's experience after completing an educational activity

using DICTs. For this stage, it is suggested to apply the focus group technique [2] to allow learners to answer open questions and report on their individual and collective experiences during the educational activity, as well as allow learners to reflect on and self-assess the learner's experience (Figure 4), and to have the freedom to be active and critical subjects in their learning, as guided by Lykke et al. [11]. For the third stage, it is essential to have completed the during-evaluation checklists the activity, both for the teacher and the learner. It is also necessary for all learners to have completed the activity proposed by the teacher. For this stage, it is suggested that the teacher act as a moderator during the discussions with the learners. Learners can express themselves randomly and everyone doesn't need to answer all the questions. The current full version of LEEM is available at the following link¹.

At each stage of LEEM, different scales are used to capture different aspects of LX. For the pre-evaluation checklist, the scale of pairs of opposing adjectives was used as it allows a gradation with 5 points and because it is possible to outline the profile of the learners. For the during-evaluation checklist, the SAM scale was used to record feelings. This scale allows exploring the affective learning experiences of learners during an educational activity. The 5-point Likert scale was also used to record learning as it allowed learners to self-reflect on their learning experiences. And for the post-evaluation checklist, open questions were used to allow learners to answer open questions and report on their individual and collective experiences during the educational activity.

Have you ever experienced a situation similar to the activity developed? Comment.

What was your contribution to the activity? Comment.

Figure 4: Part of the LEEM post-evaluation

4 CASE STUDY

An evaluative case study was carried out as suggested by Yin [27], as this type of study makes it possible to investigate a phenomenon in a real-world context, taking into account different variables of interest and multiple sources of evidence, as well as making it possible to collect and analyze the data. Thus, this case study was carried out to evaluate LX and its possible improvements through the use of LEEM in a real context. This study was approved by the research ethics committee at the researcher's institution.

Population and Sample. The LEEM case study was carried out with learners from the CS & BI courses who were coursed Software Engineering (SE) at UFPR, and with the respective teacher of this course. The teacher was invited by email to participate of the study. She agreed to take part voluntarily and was given the guidelines of the study. Next, the teacher, together with the researchers and authors of this paper, invited the learners to participate in the study in one of the in-person classes, explaining the purpose of the study.

¹<https://figshare.com/articles/thesis/Case Study of LEEM/24121398?file=42316878>

We presented the data collection instrument to the learners who expressed an interest in taking part in the study.

Context. The practical educational activity in which LEEM was used to assess LX was the last practical activity of the SE course. The goal of this practical activity was to carry out a software development plan, considering one of the software development methodologies already predefined in a previous activity of the same course. This activity was organized into two stages, (1) the preparation of a report and (2) its presentation. The proposal for this educational activity was presented to the learners on May 31, 2023, together with the LEEM. Regarding the formation of the groups, the teacher instructed that each group should have a maximum of three members, and preferably should be kept the same group from the previous activity.

To prepare the report, the teacher instructed each group to have the same software project manager as in the previous activity. The teacher also instructed the learners to use the material from Wazlawick [26]; to start by defining the structure (the items) of the project management plan; and to identify the information needed to fill in this plan. If the learner noticed that any information was missing, the teacher advised them to ask the project manager during the lesson and suggested as an example that the duration of the project should be 11 months. To complement the report, the teacher left it optional to carry out project planning with estimation. The teacher also instructed the learners to define the Story Points for the User Stories on a web page². She also suggested further reading, such as: Paiva et al. [16]; Prasada Rao et al. [18]; Cohn [3]; Knaster and Leffingwell [9]; Porru et al. [17]; and Schank and Berman [20]. The teacher gave the learners the freedom to choose the DICTs to do this activity. The final version of the report was due on June 21, 2023. In addition, two classes were made available for the presentation of this educational activity, with time limits of a minimum of 5 minutes and a maximum of 7 minutes per group.

Instruments. For this case study, the teacher and the learners first signature of the Informed Consent Form (ICF) when they agreed to take part in the study. After signing the ICF, both the teacher and the learners answered the characterization questionnaire. Afterward, the teacher asked the learners to complete the Pre-evaluation checklist (1st stage). The teacher then taught her lesson as normal, as stipulated in the syllabus. Then, as provided for in the LEEM model, the During-evaluation was applied after completing 50% of the educational activity that the teacher had taught in the lesson. In this 2nd stage, the teacher instructed the learners to complete the during-evaluation checklist, and the teacher also completed her during-evaluation checklist. At the end of the activity (3rd stage), the teacher asked the learners to answer the questions set on the post-evaluation checklist. Instruments are available on Link³.

Evaluation preparation and execution. Before carrying out the study, a presentation was given to the participants, in which they were shown the study instruments and the step-by-step procedure to be carried out as detailed above. This presentation lasted around 30 minutes. At the end of the presentation, the researcher took questions from the participants about the topics covered. After this

meeting with the participants, the instructions were sent by e-mail containing the link to the data collection instruments. The participants carried out the activities and their answers were stored on the Survey Monkey⁴ platform. In short, 23 learners completed the following stages of the study. The learners who agreed to take part in the study: (1) they signed ICF; (2) they completed the characterization questionnaire; (3) they answered the LEEM pre-evaluation checklist; (4) they completed 50% of the educational activity; (5) they answered the during-evaluation checklist the LEEM; (6) they completed the other 50% of the educational activity; and (7) they answered the LEEM post-evaluation checklist. The teacher carried out the following stages of the study: (1) the teacher, who agreed to take part in the study, signed the ICF; (2) she filled in the characterization questionnaire; (3) she proposed an educational activity in the form of practical activity to her class; (4) after the learners had carried out 50% of the educational activity, the teacher answered the LEEM during-evaluation checklist (teacher); and (5) after the learners had finished the educational activity, the teacher watched each group's presentation and evaluated them. In this context, the teacher was responsible for accompanying his learners in responding to the LEEM checklist during the educational activity. In addition, she had the freedom to define the 50% of the educational activities to guide the learners in answering the LEEM during-evaluation checklist. The researchers did not influence in this decision.

Data Analysis. The data obtained from the LEEM evaluation questionnaires was analyzed quantitatively and qualitatively. The quantitative data was analyzed using descriptive statistics, looking at the participants' responses about word pairs, the SAM, and Likert scales used in the LEEM pre-evaluation and during-evaluation stages. Before the qualitative analysis, the data was cleaned, coded, and then organized. To code and organize the qualitative data, we followed a subset of the steps of the Grounded Theory (GT) method, defined by Service [21]. GT has three stages in the coding process: open coding (1), axial coding (2), and selective coding (3). The first stage is open coding, in which the data is coded according to the answer given by each participant in the LEEM post-evaluation stage. Subsequently, in axial coding, the codes were grouped according to their properties and related to each other, thus forming categories that represent their characteristics. Selective coding was not carried out, as the intention was not yet to create a theory. This is because the open and axial coding stages were sufficient to understand LX in the activity of the SE course through the use of LEEM.

5 RESULTS

Regarding the **characterization of the learners' profile**, gender was collected, where it was noted that 78% (N = 18) are male, 13% (N = 3) are female and 9% (N = 2) did not want to identify themselves. Regarding the age of the learners, they are between 20 and 30 years old. As for looking for new DICTs to support learning, in addition to those suggested by the teachers, 52% (N = 12) of the learners usually look for new DICTs, and 48% (N = 11) of the learners don't usually look for new DICTs. Regarding whether LX has ever been evaluated in any course, 92% (N = 22) of the learners answered "no" and only one learner answered "yes". The learner who said "yes" stated that LX had been evaluated in another course through open

²<https://getstream.io/blog/user-story-examples/>

³<https://figshare.com/account/articles/24121398>

⁴<https://www.surveymonkey.com>

questions asked by the teacher. Regarding how long they usually use TDICs per day to support learning, 45% (N = 10) of learners use between 1h and 2h daily, 32% (N = 7) of learners use less than half an hour daily, 23% (N = 5) of learners use between 2h and 4h per day and 5% (N = 1) of learners use more than 6h per day. Regarding using TDICs for which learning activities, 76% (N = 16) of learners use them for preparing and presenting work, 24% (N = 5) of learners use them for projects and seminars, 19% (N = 4) of learners use it to fix content, 19% (N = 4) of learners use it to study for tests and 5% (N = 1) of learners use it for mental maps. Regarding which TDICs the learner usually uses, 88% (N = 21) use *Google Drive*, 83% (N = 20) use *GitHub*, 63% (N = 15) use *Overleaf*, 29% (N = 7) use *Trello*, 25% (N = 6) use *Canva*, 13% (N = 3) use *Notion*, 8% (N = 2) use *Figma* and the rest correspond to 4% (N = 1) each, being *Astah*, *Draw.io*, *CodiMd*, *Quire*, *Moodle C3SL*, *UFPR Virtual*, *Pandoc* and *chatGPT*.

Regarding **characterization of the teacher**, she is female and between 51 and 60 years old. She has used DICTs to support teaching and learning since the beginning of her teaching career. When asked if she had ever evaluated LX in any of her courses, the teacher said yes and reported that she had done so in the Introduction to Computer Science course during the pandemic and in the SE course using Google Forms. The results obtained through LEEM checklists are presented below.

The LEEM pre-evaluation checklist contains 12 sentences with opposing word pairs (Figure 5). For analytical purposes, the answers given to the sentences on this checklist were categorized on a five-point scale, ranging from 0 to 20, 21 to 40, 41 to 60, 61 to 80, and 81 to 100 between the word pairs, just as Lykke et al. [11] did in her study. For example, for the word pair, “Optimistic/Pessimistic” the answers from 0 to 40 represent being more “Optimistic”, the answers from 41 to 60 represent neutral for the word pair “Optimistic/Pessimistic” and the answers from 61 to 100 represent being more “Pessimistic”; this same organization is configured for the other word pairs. The results of the pre-evaluation checklist show that for the word pair “act or react” 48% (N = 11) of the learners answered 41 to 60 (neutral answers), while for the word pair, “participative or shy” 52% (N = 12) of the learners answered 61 to 80 (shyer answers). This shows that more than half of the learners considered themselves shy. In addition, these learners considered themselves to be neutral when it came to acting and/or reacting. It is believed that by identifying these characteristics of the learners, if the teacher wished to use this information to form groups, he or she could better balance these groups so that there were not only shy learners in the same group, for example. The data obtained from the pre-evaluation checklist can contribute to assigning learners to each group. However, it’s worth pointing out that it’s up to the teacher to decide whether to use the data from this checklist to form groups or not. They can also use this data to get to know their learners better and promote better experiences. For the word pair “contribute or not impact” 74% (N = 17) of the learners answered 0 to 20 (answers closer to contribute) and for the word pair “encouraging or discouraging”, 86% (N = 20) of the learners answered 0 to 40 (answers closer to encouraging). Therefore, it is observed that the majority of learners believe that DICTs can encourage, contribute and thus provide more meaningful experiences. To reiterate, in the word pair “ease or difficulty”, 52% (N = 12) of the learners answered 0 to 20 (answers closer to ease). It is noted that learners find it easy

to use DICTs, as they are already part of everyday life in general. For the word pair “true or uncertain”, 43% (N = 10) of the learners answered 0 to 20 (answers closer to true). It is believed that learners consider their true ideas, as they authentically seek to contribute to educational activities.

The during-evaluation checklist (learner) contains sentences from Q1 to Q7 (Figure 6) with a 9-point scale SAM and from Q8 to Q23 (Figure 7) with a 5-point Likert scale. To better visualize the results, the SAM scale was organized into 3 points, where sad responses are those marked to the left of the central column (1 to 3); neither sad nor happy responses are those marked in the central column (4 to 6); and happy responses are those marked to the right of the central column (7 to 9), just as Souza et al. [24] adopted in their study. The results of the evaluation checklist for the SAM scale (Figure 6) show that 17.39% (N = 4) learners answered sadly when asked how they felt emotionally on the day they answered the checklist (Q1). It is believed that these few participants declared themselves sad, because the educational activity they took part in was already the last of the course, not generating enthusiasm or any feeling of happiness. On the other hand, 34.79% (N = 8) learners answered happy, that is, they were feeling happy on the day of answering the checklist. Regarding how the learners felt when looking for support material (Q5), 17.39% (N = 4) learners answered sadly. In this sense, it is believed that these learners did not feel enthusiastic about looking for new content or materials to carry out the educational activity. Regarding how learners felt about working collectively (Q6), 8.69% (N = 2) learners answered sadly. This may have been because working collectively on an educational activity has its challenges. Regarding how they felt when they had to make the improvements requested by the teacher (Q7), 4.34% (N = 1) learners answered sadly. This may have been because they thought that the changes requested would be too much work to make.

Also for the during-evaluation checklist (learner), about the sentences answered using the Likert scale (Figure 7), the results show that the participants answered totally disagree for Q10 (8.69% | N = 2), Q16 (4.34% | N = 1), Q18 (4.34% | N = 1) and Q23 (4.34% | N = 1). It is believed that the participants responded totally disagree, as they were not interested in looking by DICTs (Q10), due to lack of time or because they already knew some tools that could help them in this activity. Another learner didn’t feel that they had learned all the content of the educational activity (Q16), perhaps because they had difficulty carrying out the activity. One learner also felt that didn’t understand the content of the educational activity using DICTs (Q18). It is believed that he learned through texts and other printed reference materials. Finally, another learner had the feeling that he didn’t persist in the face of obstacles that arose during the educational activity (Q23). In other words, it is believed that this learner found it difficult to complete the activity, possibly giving up.

The during-evaluation checklist (teacher) has 17 sentences answered on a 5-point Likert scale, ranging from Strongly Disagree to Strongly Agree. The teacher did not disagree with any of the sentences. She partially agreed with the sentences: they liked working collectively (Q3), they liked the result of the educational activity (Q4), they found the DICTs suitable (efficient) (Q6), they found the DICTs sufficient (effective) (Q7), they thought the DICTs applied well to the context (Q8), the DICTs met the learners’ needs

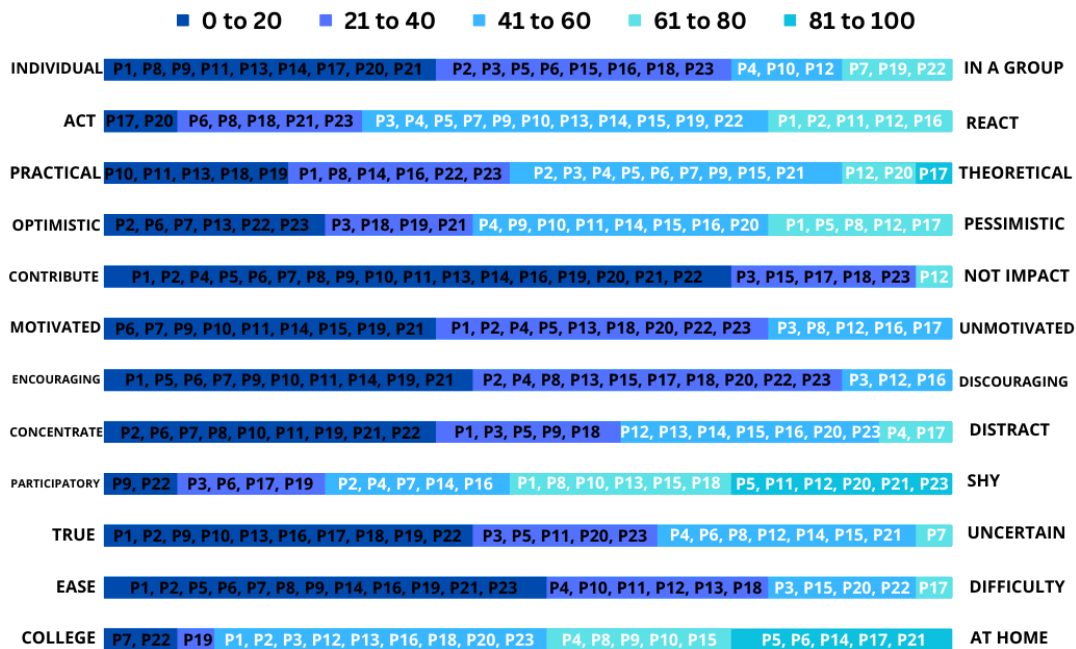


Figure 5: Results of the LEEM pre-evaluation checklist

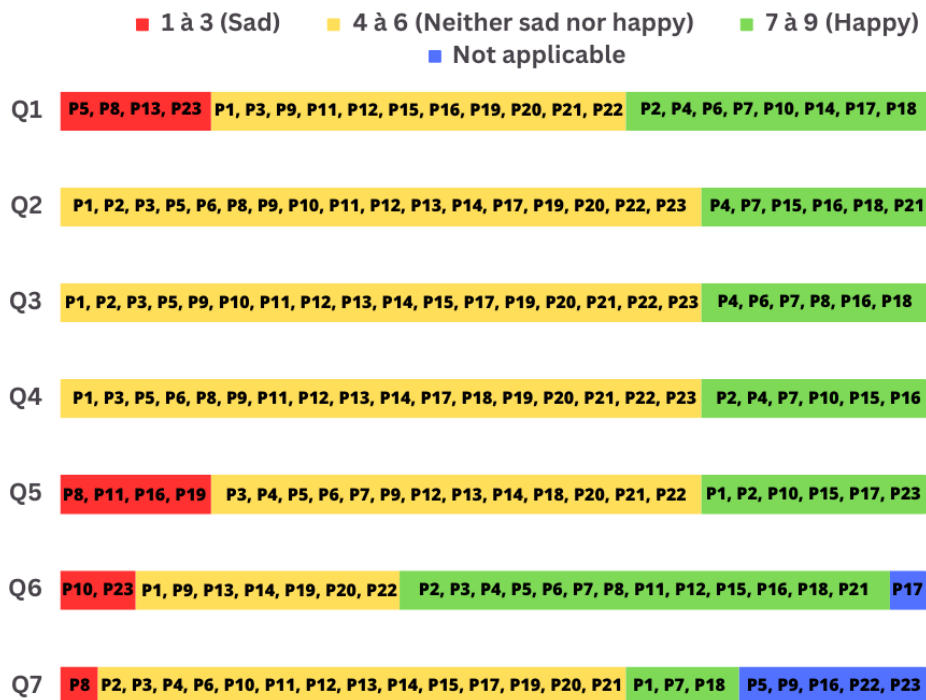


Figure 6: Results of the LEEM during-evaluation checklist (learner)

(Q9), they learned the content in its entirety (Q10), they understood the content using DICTs resources (Q12), they understood the content when talking to colleagues in the group (Q13) and they

persisted (didn't give up) in the face of obstacles (Q17). It is believed that the teacher's perception when observing the learners

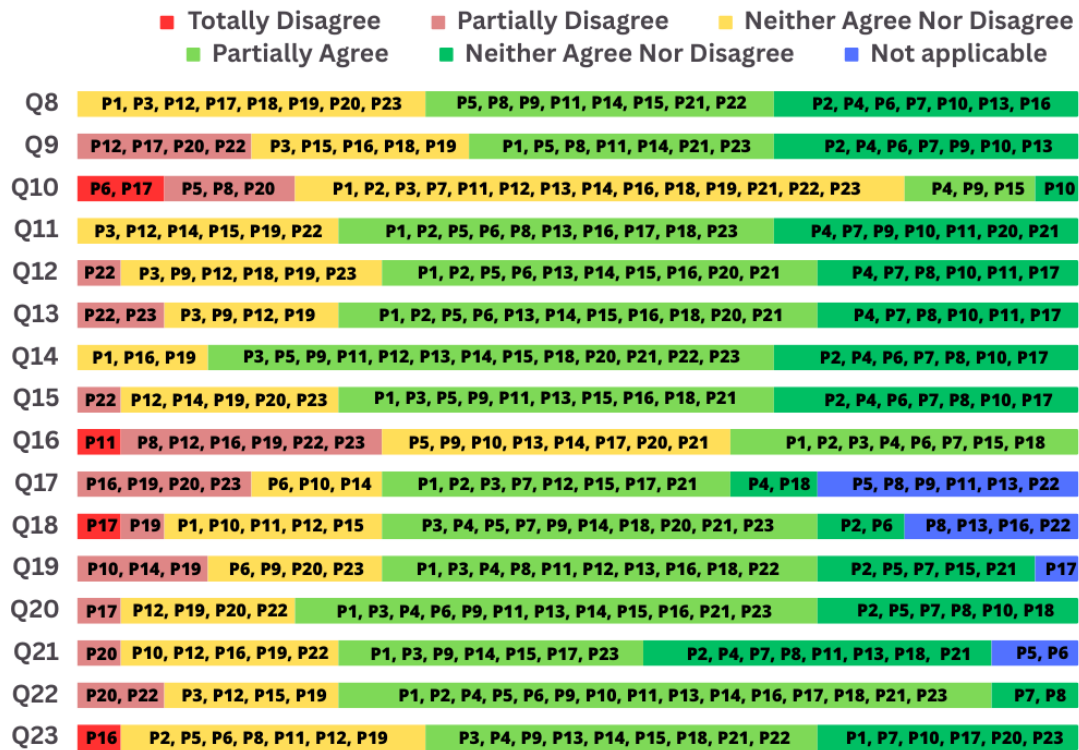


Figure 7: Results of the LEEM during-evaluation checklist (learner)

carrying out the educational activity is consistent with the learners' self-assessments. For example, the majority of learners in the evaluation checklist enjoyed working collectively (Q6), but some reported that they didn't have a good experience. This was also perceived by the teacher, through her answer to Q3, where she partially agreed that the learners enjoyed working collectively. In all other questions in the evaluation checklist, the teacher totally agreed with the sentences.

In the post-evaluation, there is a set of 11 open questions, which were analyzed qualitatively. This analysis aimed to evaluate the learners' LX after completing the educational activity. For this analysis, the tool Atlas.ti version 7⁵ was used. The categories and their respective codes identified in this analysis are presented below. For the **general comments on the experience of the activity** category, learner P10 reported: "at the beginning (of the educational activity) I wasn't that interested, but, as I learned from the materials available, my interest increased". Because it was a collaborative activity, P9 said that "it was quite difficult to find time when everyone could work together and clear doubts about conflicts. In the end, some members participated more". P11 reported that his contribution to the activity was necessary for it to be completed. It is believed that as this was an educational activity that was being carried out at the end of the course, this had an impact on the dedication and experience of the learners, which is confirmed by P18's report that the learner "was busy with many other things", and so was unable to devote himself fully to this educational activity. On the other hand,

learner P10 points out that he hadn't done an educational activity like this before and that "it had a positive impact" on his experience.

For the **general comments about the activity itself** category, learners report on their experiences during the educational activity. P10 said that "it was necessary to think about all the possible risks and costs that a real project might have". P20 said that "from his own experience with work, he usually ends up not having all these steps in the development methods". In addition, P9 said: "[the activity] generated challenges, because I had to think about possible divisions for iterations without having the use cases". P20 also pointed out that "getting to know new tools may come to be useful at some point". P13 reported: "there were deadlines and people whose grade depends on my doing my part of the activity".

For the **participation of members in the activity** category, the learners reported that "everyone [in the group] was proactive" (P3), "everyone contributed to the development of the activity" (P4), "everyone was participative" (P7), "everyone participated in the project with activities and relevant research" (P15), "we worked together" (P21), there were "online meetings with everyone present" (P7), "everyone collaborated on part of the task" (P21). There were also reports that "there was more participation from some members" (P9) and "one member of the group didn't participate and didn't give any feedback" (P10). It is believed that most of the participants were able to work as a group and had a good experience in carrying out the educational activity, except for P17 who preferred to do the activity alone.

⁵<https://atlasti.com/>

6 DISCUSSIONS

Through this case study, it can be said that the evaluation of LX is an essential component for verifying learners' experiences in the use of DICTs. In this sense, Tabares et al. [25], in an evaluation of LX that took place during an educational activity, highlighted that the web application contributed to diagnosing weaknesses in learning, making it possible to identify behaviors, preferences and even some difficulties of the learners. In the study by Magyar and Haley [12], the LX assessment only took place after the educational activity. It is believed that because the LX was only evaluated after the activity had been completed, the use of the Gallery Tool demotivated the learners, as several of the expected functionalities were not present. If an evaluation had taken place during the educational activity, this type of issue would have been identified. Unlike Magyar and Haley [12] and Tabares et al. [25], in our case study using the LEEM model, it was possible to follow the learners' experiences of using the DICTs both during and after the educational activity. Through the teacher's evaluation during our case study, she noticed that the learners felt comfortable using the DICTs and that they were easy to use (Q5 and Q14). This was confirmed by the feedback given by the learners themselves in their evaluations during the activity (Q11 and Q20) and in the post-evaluation, such as the feedback given by P7 who said that *"(...) the DICTs helped with the development, [such as] meetings, research..."*.

We would also like to emphasize that one of the results of the LEEM pre-evaluation was that P10 believed that DICTs could help him concentrate when carrying out an educational activity (Figure 5). In turn, for the learner during-evaluation checklist, this same learner reported that he preferred to use DICTs to carry out the educational activity (Q8) (Figure 7). Finally, in the post-evaluation, he also reported that the educational activity with DICTs is interesting. He also pointed out that even though he wasn't interested at the beginning of the activity, as he learned with the materials provided and using the available DICTs, his interest grew.

In short, the main experiences identified in LEEM, through this study, is the monitoring of the data obtained through the LEEM stages (pre, during, and post), as detailed in Section 5. This shows that there were learners who were not interested at the start of the educational activity, but as the experience progressed, their interests changed. In addition, because it was a collaborative activity, it was possible to see the importance of communication and responsibility, where the experience was different for most of the learners, for example, some didn't work, some worked well, and one learner decided to do the activity alone. Therefore, answering the research question of this study, it can be said that the goal of LEEM was achieved, because through it the evaluation of LX was carried out throughout the educational activity. In addition, it was observed that the LEEM proved to be useful in supporting the improvement of learners' experiences in the use of DICTs in educational activities.

7 CONCLUSION AND FUTURE WORK

This paper aims to present LEEM, a model that allows LX to be assessed and improved when interacting with DICTs, encompassing different forms of evaluation and different elements of LX. A case study was also carried out to evaluate LX in a real context with learners and their teacher. Through the results obtained with the

LEEM, the teacher and learners were able to evaluate the LX and identify points for improvement in the educational activity carried out. The data obtained through the LEEM can also be used so that the teacher can rethink her teaching strategies when she notices the negative experiences reported by the learners.

Through the analysis of this case study, it was noted that for the pre-evaluation checklist, one learner reported that he prefers to work individually, and in the post-evaluation, he reported that he worked in a group, where all the members of the group were participative. Another learner also said that he preferred to work individually in the pre-evaluation checklist, and reported in the post-evaluation that he worked alone, highlighting that he felt motivated to carry out the educational activity anyway. In addition, another learner reported in the pre-evaluation that he prefers to work in a group and reported in the post-evaluation that a colleague in the group did not collaborate with the educational activity. This shows that we have obtained different reports with different learning experiences for the same educational activity. And that, even though the expectation reported by the learner in the pre-evaluation was one, what happened during the educational activity affected their experience, as they reported in the post-evaluation. It also highlights the importance of the teacher instructing and accompanying the learners throughout the educational activity so that the LX can be evaluated and improved, providing a more engaging and memorable experience for learners.

Some limitations were considered for this study. One of them may have been the noisy environment (classroom) in which the LEEM was being used. In addition, interruptions during the use of the LEEM may also have occurred, either by the students themselves during the lesson. Another limitation is the veracity of the information obtained because as with any type of evaluation carried out with learners and teachers, the reliability and veracity of the information depends directly on who is answering. In addition, another limitation may have been that learners were given the freedom to form their groups. The data from the pre-evaluation checklist could have been used to support this group formation by the teacher, seeking to balance the profiles of the learners in each group. However, we did not want to interfere in this group formation process. Another limitation considered is the selection of participants of the study, as it was carried out with SE learners, and there was no representation of other academic disciplines. This may make it difficult to generalize the results, as there is no diversity of perspectives and experiences from different academics. Finally, another limitation is the fact that no DICTs were pre-defined for the educational activity since the learners chose the ones they wanted to use in this activity. Therefore, based on these limitations, these results are considered to be indications and not conclusions about the use of LEEM in a real context.

In future work, we intend to evaluate the acceptance of the learners and the teacher about the use of LEEM. It is intended to carry out more studies considering a more varied sample and covering a wider range of academic backgrounds, besides varying class schedules across different majors. Moreover, we desire to deepen the collection of teacher feedback with questionnaires or interviews to gather their perspectives on the effectiveness of the LEEM and its practical applicability.

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REFERENCES

- [1] Briana Bettin. 2023. Challenges, Choice, & Change: Experiences and Reflections from the First Semester of a Technology and Human Futures Course. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (<conf-loc>, <city>Toronto ON</city>, <country>Canada</country>, </conf-loc>) (SIGCSE 2023). Association for Computing Machinery, New York, NY, USA, 235–241. <https://doi.org/10.1145/3545945.3569872>
- [2] Jared R Chapman, Eugene L Seeley, Norman S Wright, Lowell M Glenn, and Lynn L Adams. 2016. An empirical evaluation of a broad ranging e-text adoption with recommendations for improving deployment success for students. *e-Journal of Business Education and Scholarship of Teaching* 10, 2 (2016), 1–14. <https://eric.ed.gov/?id=EJ1167339>
- [3] M. Cohn. 2005. *Agile Estimating and Planning*. Pearson Education, USA. <https://books.google.com.br/books?id=BuFWHfRJsC>
- [4] Claiton Marques Correa, Gabriel Viegas Maciel de Freitas, André Luis dos Santos Eberhardt, and Milene Selbach Silveira. 2021. From Now on: Experiences from User-Based Research in Remote Settings. In *Proceedings of the XX Brazilian Symposium on Human Factors in Computing Systems* (Virtual Event, Brazil) (IHC '21). Association for Computing Machinery, New York, NY, USA, Article 25, 7 pages. <https://doi.org/10.1145/3472301.3484334>
- [5] Ronghuai Huang, J Michael Spector, and Junfeng Yang. 2019. *Educational Technology a Primer for the 21st Century*. Springer, Singapore. <https://doi.org/10.1007/978-981-13-6643-7>
- [6] Rafael Eiki Matheus Imamura and Maria Cecília Calani Baranauskas. 2019. A Framework for Socio-Enactive Educational Systems: Linking Learning, Design, and Technology. In *IHC '19: XVIII Brazilian Symposium on Human Factors in Computing Systems* (Vitória, Espírito Santo, Brazil) (IHC '19). Association for Computing Machinery, New York, NY, USA, Article 1, 11 pages. <https://doi.org/10.1145/3357155.3358443>
- [7] ISO9241-210. 2019. Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems. <https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:v1:en> [Online; acessado em 23/02/2022].
- [8] Atsuko Kawano, Yuji Motoyama, and Mikio Aoyama. 2019. A LX (Learner EXperience)-Based Evaluation Method of the Education and Training Programs for Professional Software Engineers. In *Proceedings of the 2019 7th International Conference on Information and Education Technology* (Aizu-Wakamatsu, Japan) (ICIET 2019). Association for Computing Machinery, New York, NY, USA, 151–159. <https://doi.org/10.1145/3323771.3323789>
- [9] Richard Knaster and Dean Leffingwell. 2020. *SAFe 5.0 distilled: achieving business agility with the scaled agile framework*. Addison-Wesley Professional, Califórnia, USA. [https://books.google.com.br/books?hl=pt-BR&lr=&id=mtnPAAAAQBAJ&oi=fnd&pg=PT9&dq=Fibonacci+scale+\(agile\)&ots=9VFNZdFd3M&sig=CSV0Ie-gX0-UikCgVGjgmLHMw#v=onepage&q&f=false](https://books.google.com.br/books?hl=pt-BR&lr=&id=mtnPAAAAQBAJ&oi=fnd&pg=PT9&dq=Fibonacci+scale+(agile)&ots=9VFNZdFd3M&sig=CSV0Ie-gX0-UikCgVGjgmLHMw#v=onepage&q&f=false)
- [10] Peter Lang. 1980. Behavioral treatment and bio-behavioral assessment: Computer applications. , 119–137 pages. <https://cir.nii.ac.jp/crid/1574231875645651456>
- [11] Marianne Lykke, Mayela Coto, Christian Jantzen, Sonia Mora, and Niels Vandel. 2015. Motivating Students through Positive Learning Experiences: A Comparison of Three Learning Designs for Computer Programming Courses. *Journal of Problem Based Learning in Higher Education* 3, 2 (2015), 80–108. <https://doi.org/10.5278/ojs.jpblhe.v0i0.1130>
- [12] Nathan Magyar and Stephanie R. Haley. 2020. Balancing Learner Experience and User Experience in a Peer Feedback Web Application for MOOCs. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3375232>
- [13] Suellen R. Martinelli and Luciana A. M. Zaina. 2021. Learning HCI from a Virtual Flipped Classroom: Improving the Students' Experience in Times of COVID-19. In *Proceedings of the XX Brazilian Symposium on Human Factors in Computing Systems* (Virtual Event, Brazil) (IHC '21). Association for Computing Machinery, New York, NY, USA, Article 34, 11 pages. <https://doi.org/10.1145/3472301.3484326>
- [14] Luã Marcelo Muriana and Maria Cecília Calani Baranauskas. 2021. Affecting User's Self-Esteem: Analysis under the Self-Determination Theory Perspective and Design Recommendations. In *Proceedings of the XX Brazilian Symposium on Human Factors in Computing Systems* (Virtual Event, Brazil) (IHC '21). Association for Computing Machinery, New York, NY, USA, Article 1, 12 pages. <https://doi.org/10.1145/3472301.3484331>
- [15] Eeva Nygren, A Seugnet Blignaut, Verona Leendertz, and Erkki Sutinen. 2019. Quantizing affective data as project evaluation on the use of a mathematics mobile game and intelligent tutoring system. *Informatics in Education* 18, 2 (2019), 375–402. <https://www.ceeol.com/search/article-detail?id=804185>
- [16] Edgy Paiva, Danielly Barbosa, Roberto Lima Jr, and Adriano Albuquerque. 2010. Factors that influence the productivity of software developers in a developer view. In *Innovations in computing sciences and software engineering*. Springer, Heidelberg, 99–104. https://doi.org/10.1007/978-90-481-9112-3_17
- [17] Simone Porru, Alessandro Murgia, Serge Demeyer, Michele Marchesi, and Roberto Tonelli. 2016. Estimating Story Points from Issue Reports. In *Proceedings of the The 12th International Conference on Predictive Models and Data Analytics in Software Engineering* (Ciudad Real, Spain) (PROMISE 2016). Association for Computing Machinery, New York, NY, USA, Article 2, 10 pages. <https://doi.org/10.1145/2972958.2972959>
- [18] Ch Prasad Rao, P Siva Kumar, S Rama Sree, and J Devi. 2018. *An agile effort estimation based on story points using machine learning techniques*. Vol. 712. Springer, Singapore. 209–219 pages. https://doi.org/10.1007/978-981-10-8228-3_20
- [19] Jenny Ruiz and Monique Snoeck. 2018. Adapting Kirkpatrick's Evaluation Model to Technology Enhanced Learning. In *MODELS '18: ACM/IEEE 21th International Conference on Model Driven Engineering Languages and Systems* (Copenhagen, Denmark) (MODELS '18). Association for Computing Machinery, New York, NY, USA, 135–142. <https://doi.org/10.1145/3270112.3270114>
- [20] Roger C Schank and Tamara R Berman. 2003. The Pervasive Role of Stories in Knowledge and Action. , 287–313 pages. <https://books.google.com.br/books?hl=pt-BR&lr=&id=Idl4AgAAQBAJ&oi=fnd&pg=PT297&dq=12+The+Pervasive+Role+of+Stories+in+Knowledge+and+Action&ots=-4u4tIVorj&sig=h9AgOyVMCQ75RcZGEJs1e-EVio#v=onepage&q=12%20The%20Pervasive%20Role%20of%20Stories%20in%20Knowledge%20and%20Action&f=false>
- [21] Robert W. Service. 2009. Book Review: Corbin, J., & Strauss, A. (2008). Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory (3rd ed.). Thousand Oaks, CA: Sage. *Organizational Research Methods* 12, 3 (2009), 614–617. <https://doi.org/10.1177/1094428108324514>
- [22] Lei Shi. 2014. Defining and Evaluating Learner Experience for Social Adaptive E-Learning. In *2014 Imperial College Computing Student Workshop (OpenAccess Series in Informatics (OASIs), Vol. 43)*, Rumyana Neykova and Nicholas Ng (Eds.). Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, Dagstuhl, Germany, 74–82. <https://doi.org/10.4230/OASIs.ICCSW.2014.74>
- [23] Elliot Soloway, Mark Guzdial, and Kenneth E. Hay. 1994. Learner-Centered Design: The Challenge for HCI in the 21st Century. *Interactions* 1, 2 (apr 1994), 36–48. <https://doi.org/10.1145/174809.174813>
- [24] Dyego Souza, Jarbele Coutinho, and Reudismam de Sousa. 2022. LoopAcademic: Helping the Learning and Teaching of Introductory Programming. *Perspectives and Trends in Education and Technology* 256 (2022), 561–571. https://doi.org/10.1007/978-981-16-5063-5_47
- [25] Marta S. Tabares, Paola Vallejo, Alex Montoya, Jose Sanchez, and Daniel Correa. 2021. SECA: A Feedback Rules Model in a Ubiquitous Microlearning Context. In *DATA'21: International Conference on Data Science, E-learning and Information Systems 2021* (Ma'an, Jordan) (DATA'21). Association for Computing Machinery, New York, NY, USA, 136–142. <https://doi.org/10.1145/3460620.3460745>
- [26] Raul Wazlawick. 2019. *Engenharia de software: conceitos e práticas*. Vol. 2. Elsevier Editora Ltda., Rio de Janeiro. https://books.google.com.br/books?hl=pt-BR&lr=&id=d1qnDwAAQBAJ&oi=fnd&pg=PT2&dq=wazlawick+engenharia+de+software&ots=p_xheLlhg6&sig=2uTuSURpSevgn03XoX-KXFDHm5Y#v=onepage&q=wazlawick%20de%20software&f=false
- [27] Robert K Yin. 2014. *Case study research: design and methods*. Vol. 5. Sage, Thousand Oaks, CA. 16–18 pages. https://scholar.google.com.br/scholar?hl=pt-BR&as_sdt=0%2C5&q=Case+study+research+%3A+design+and+methods&btnG=